



January 30th 2026

PReVENT 2 BI

PRotective VENTilation to PReVENT
2(secondary) Brain Injury

<Code 2heal G>

Our study

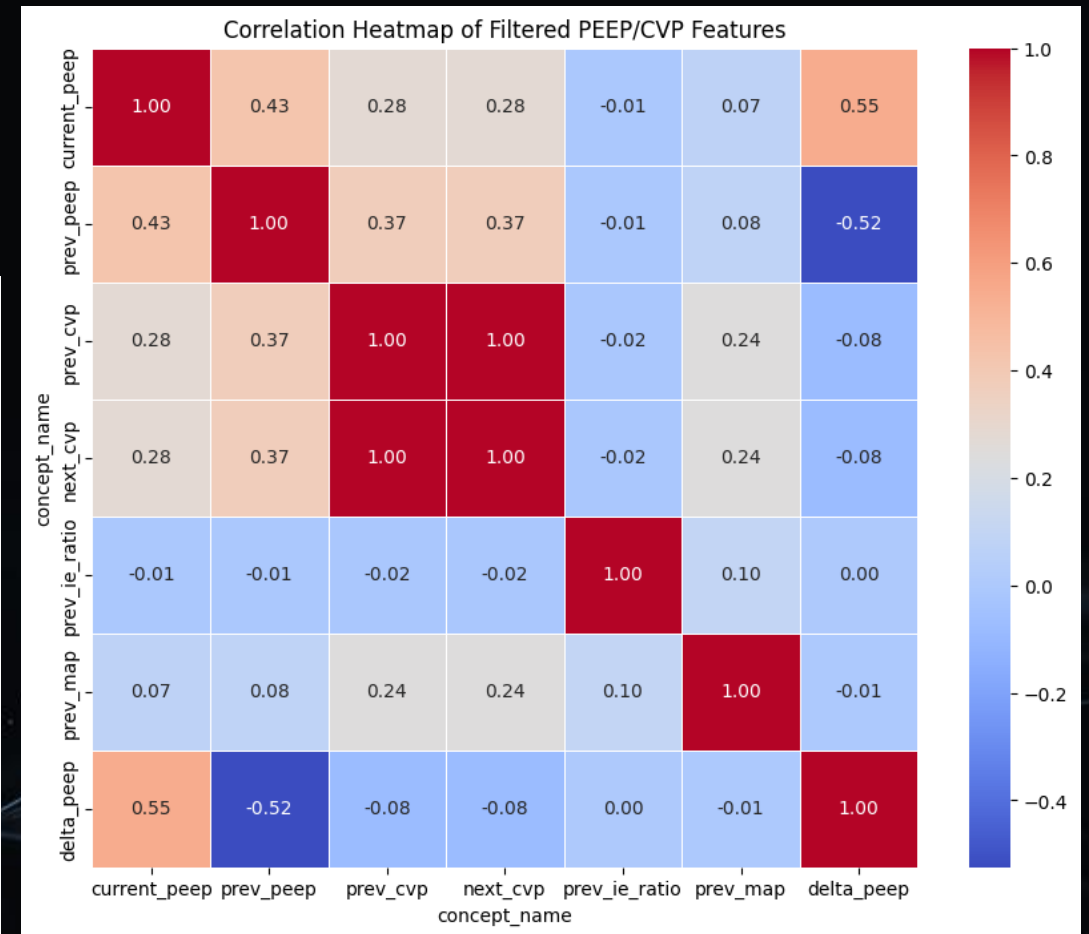
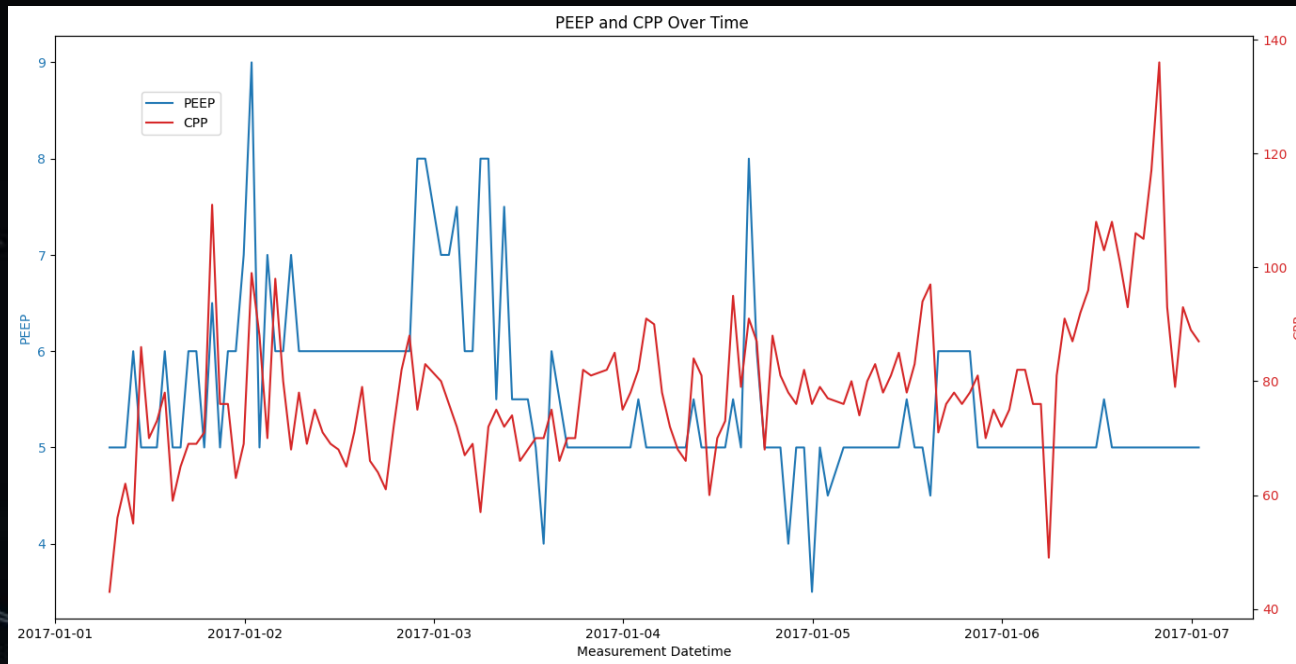
OBJECTIVE

To assess the impact of the changes in airway pressures during mechanical ventilation on cerebral perfusion pressure (CPP).

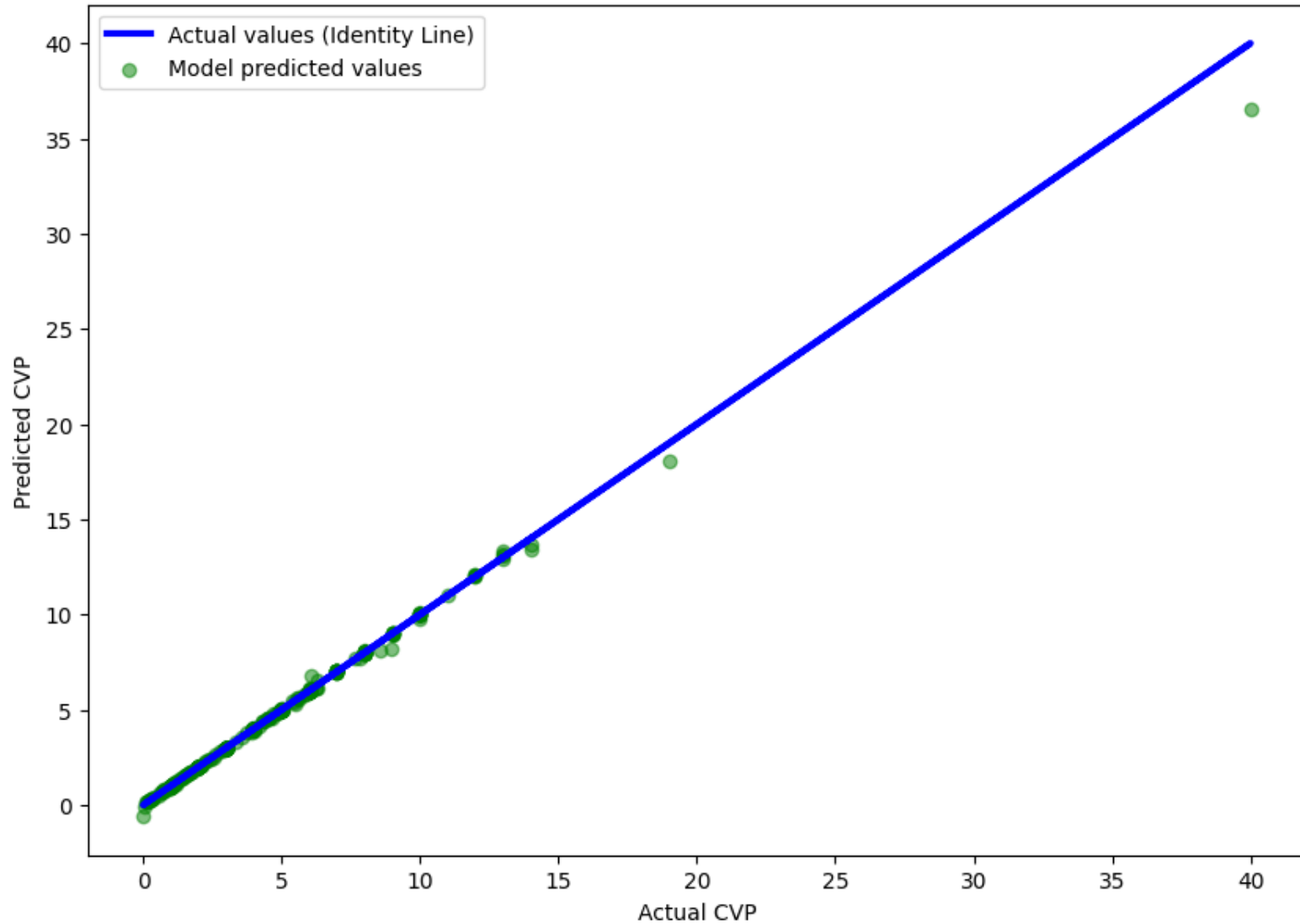
METHODS:

-  **XGBR**
-  **CNN + LSTM**
-  **Ensemble Model:**
 -  XGBoost
 -  LightGBM
 -  CatBoost
 -  Random Forest
-  **Fuzzy Logic**

TBI patient observation



XGBR Scatter Plot: Actual vs Predicted CVP (Sample Size: 500)



Test results:

XGBR:

RMSE: 0.2068;

MAE: 0.0297;

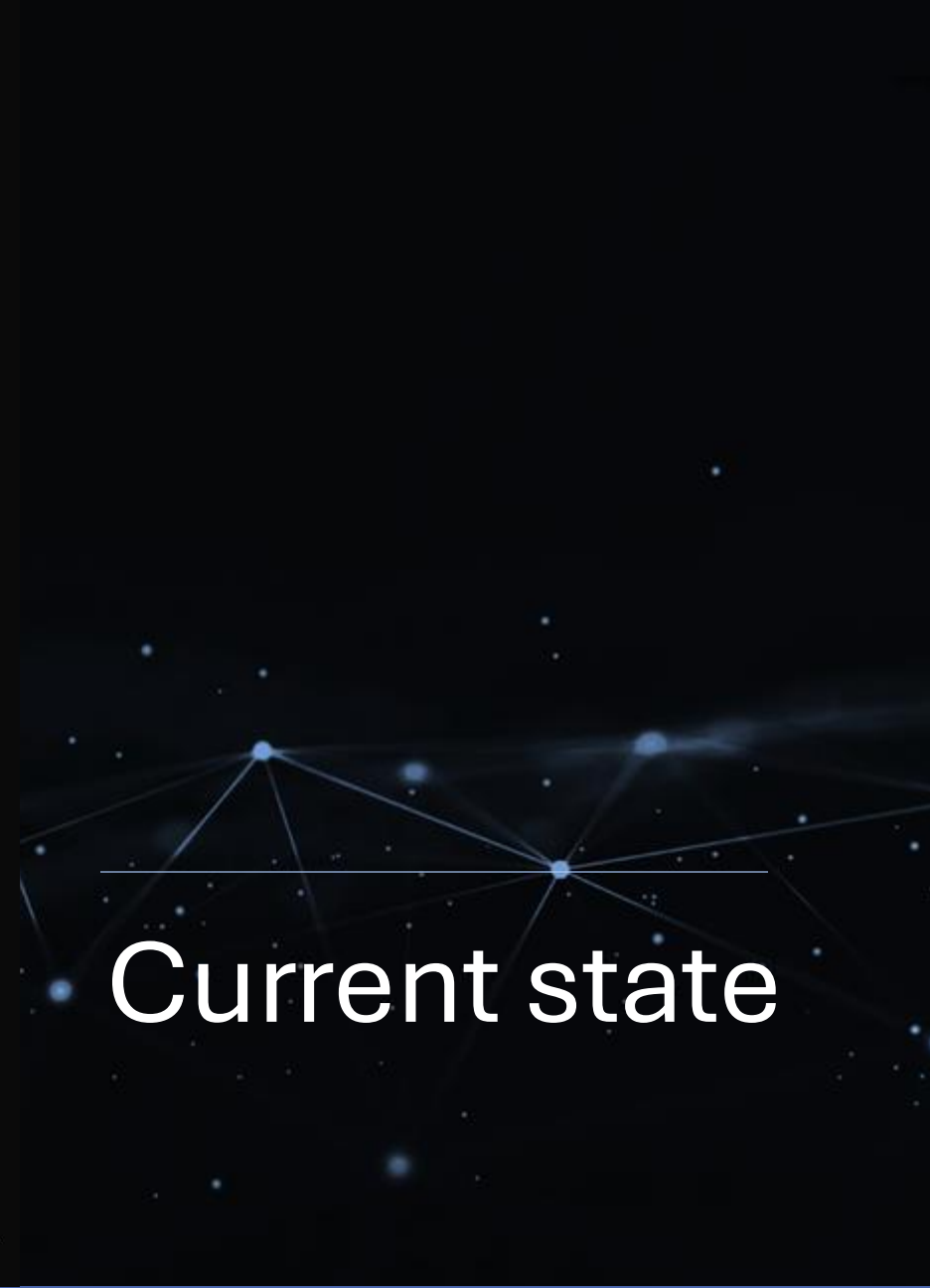
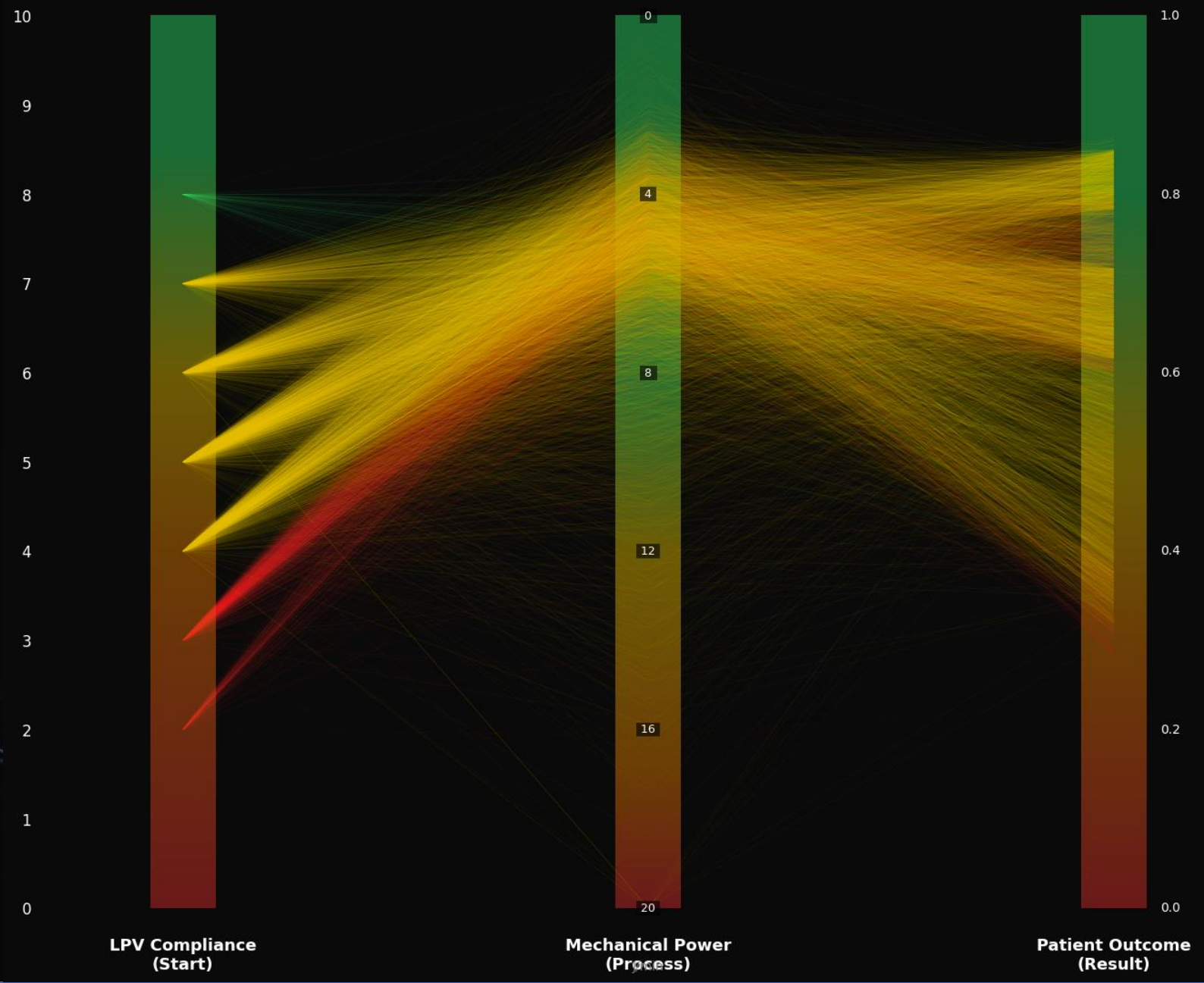
R^2 : 0.9954.

Ensemble Model:

RMSE: 0.5496;

MAE: 0.0466;

R^2 : 0.9672.



Thank you for
your attention!
